```python
print("Name: CHIRAG AGARWAL")
print("USN: 1WA24CS098")
print()

import random

class SimpleTwoRoomVacuumCleanerAgent:
    def __init__(self):
        print("\n--- Initial Room Setup ---")
        # Prompt user for initial dirt status of Room A
        while True:
            try:
                dirt_a = int(input("Enter initial dirt status for Room A
(0 for clean, 1 for dirty): "))
                if dirt_a in [0, 1]:
                    break
                else:
                    print("Invalid input. Please enter 0 or 1.")
            except ValueError:
                print("Invalid input. Please enter a number.")

        # Prompt user for initial dirt status of Room B
        while True:
            try:
                dirt_b = int(input("Enter initial dirt status for Room B
(0 for clean, 1 for dirty): "))
                if dirt_b in [0, 1]:
                    break
                else:
                    print("Invalid input. Please enter 0 or 1.")
            except ValueError:
                print("Invalid input. Please enter a number.")

        self.rooms = {'A': dirt_a, 'B': dirt_b}

        # Prompt user for initial robot location
        while True:
            initial_location = input("Enter initial robot location (A or
B): ").upper()
            if initial_location in ['A', 'B']:
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                self.current_room = initial_location
                break
            else:
                print("Invalid input. Please enter 'A' or 'B'.")

        self.actions_taken = 0
        print(f"Initial room states: {self.rooms}, Starting in room:
{self.current_room}")

    def sense(self):
        """Senses if the current room is dirty."""
        return self.rooms[self.current_room] == 1

    def suck(self):
        """Cleans the current room if it's dirty."""
        if self.sense():
            self.rooms[self.current_room] = 0
            print(f"Action: SUCK in Room {self.current_room} (Room is now
clean)")
            self.actions_taken += 1
            return True
        return False

    def move(self):
        """Moves to the other room."""
        if self.current_room == 'A':
            self.current_room = 'B'
            print("Action: MOVE RIGHT to Room B")
        else:
            self.current_room = 'A'
            print("Action: MOVE LEFT to Room A")
        self.actions_taken += 1

    def run(self, max_steps=100):
        """Runs the agent until both rooms are clean or max_steps is
reached."""
        step = 0
        while (self.rooms['A'] == 1 or self.rooms['B'] == 1) and step <
max_steps:
            print(f"\n--- Step {step + 1} ---")
```

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            print(f"Current State: Rooms={self.rooms}, Agent in
{self.current_room}")

            if self.suck():
                # If sucked, check if done or continue in same room
                pass # Agent just cleaned, will re-sense next iteration
            else:
                # If current room is clean, move to the other room
                self.move()
            step += 1

        print("\n--- Cleaning Complete ---")
        print(f"Final room states: {self.rooms}")
        if self.rooms['A'] == 0 and self.rooms['B'] == 0:
            print("Both rooms are clean!")
        else:
            print("Stopped after reaching max steps, rooms might not be
fully clean.")
        print(f"Total actions taken: {self.actions_taken}")

# Instantiate and run the agent
agent = SimpleTwoRoomVacuumCleanerAgent()
agent.run()
```

Output:

```
Name: CHIRAG AGARWAL
USN: 1WA24CS098


--- Initial Room Setup ---
Enter initial dirt status for Room A (0 for clean, 1 for dirty): 1
Enter initial dirt status for Room B (0 for clean, 1 for dirty): 0
Enter initial robot location (A or B): b
Initial room states: {'A': 1, 'B': 0}, Starting in room: B

--- Step 1 ---
Current State: Rooms={'A': 1, 'B': 0}, Agent in B
Action: MOVE LEFT to Room A

--- Step 2 ---
Current State: Rooms={'A': 1, 'B': 0}, Agent in A
Action: SUCK in Room A (Room is now clean)

--- Cleaning Complete ---
Final room states: {'A': 0, 'B': 0}
Both rooms are clean!
Total actions taken: 2
```

Vaccum Cleaner :-

1) Start
2) Initialise two rooms "A" & "B".
3) Check whether the rooms are dirty or clean.
4) Check if room is dirty

then vaccum ~~clt~~ cleaner sucks.

5) Check if room is clean

if vaccum cleaner is in room A, then move right to room B.

if vaccum cleaner is in room B, then move left to room A.

6) Stop when both rooms "A" & "B" are clean.
7) End.

Pseudo Code.

Function Reflex_Vaccum_Agent ([location, status])

returns an action

if status = dirty then return Pick the Dirt

else if location = A then return Right

else location = B then return (left)

Output for Simple Reflex Agent :

1. Initial room setup

Enter initial dirt status for room A (0 for clean, 1 for dirty) : 0

Enter initial dirt status for Room B (0 for clean, 1 for dirty) : 1

Initial robot location (A or B) : A

Initial room status : {'A' : 0, 'B' : 1},
starting in room : A

11. Step 1 →

Current State : Rooms = {'A' : 0, 'B' : 1},
Agent in A

Action : Move Right to Room B

1. Step 2 →

Current State : Rooms = {'A' : 0, 'B' : 1},
Agent in B

Action : Suck in Room B (Room is now clean)

. Cleaning Complete →

Both rooms are clean !

Total action taken = 2